

Rasul Kaireldin

Email: rkaireldin@ucmerced.edu

LinkedIn: <https://www.linkedin.com/in/rasul-kaireldin/>

Google Scholar: <https://scholar.google.com/citations?user=N8vIYW8AAAAJ&hl=en&oi=ao>

EDUCATION

- **University of California, Merced** Merced, CA
PhD, Machine Learning and Optimization; GPA: 3.86. Advisor: Miguel. A. Carreira-Perpiñán Jan. 2022 – Present
- **Nazarbayev University** Astana, Kazakhstan
Master of Science in Electrical And Computer Engineering; GPA: 3.67/4.00 (with Honors) Aug. 2018 – Jun. 2020
- **Nazarbayev University** Astana, Kazakhstan
Bachelor of Engineering in Electrical and Electronic Engineering; GPA: 3.50/4.00 Aug. 2013 – Jun. 2018

EXPERIENCE

- **University of California, Merced** Merced, CA
Graduate Research/Teaching Assistant Jan. 2022 – Present
 - **Member of machine learning research group. Research directions:** Tree Alternating Optimization; Learning Compression algorithm for finding optimally compressed neural networks.
 - **TA for the following courses:** Introduction to Machine Learning, Introduction to Artificial Intelligence, Advanced Programming, Algorithm Design and Analysis, Data Structures.
- **Meta** Bellevue, WA
PhD Software Engineer Intern at Ranking & Foundational AI team May. 2025 – Aug. 2025
 - Neural architecture search for Atlas v2 User and Ad embeddings
 - VQ-VAE based solution for user embedding diversification
- **Meta** Menlo Park, CA
PhD Software Engineer Intern in Core ML team May. 2024 – Aug. 2024
 - Improved the neural network compression for large recommendation systems on next-generation Meta Training and Inference Accelerator (MTIA).
 - Identified the best quantization strategy for sparse embedding quantization. Results were proven experimentally in terms of QPS and normalized entropy.
 - Developed quantization scheme for dynamic int8 quantization of fully connected layers in multiple models.
- **Nazarbayev University(#1 national institution in research)** Astana, Kazakhstan
Research Assistant Aug. 2018 – Jul. 2021
 - **Member of Complex Networks and Systems Laboratory. Research directions:** Deep learning approach to optimal activity control over multilayer networks; Time-series prediction of information diffusion in complex networks.
 - Developed and deployed 2 web scraping solutions for analysis processes.
 - Innovated machine learning-based algorithms for link prediction in temporal networks, resulting in a 3% improvement in accuracy.
- **Astana IT University** Astana, Kazakhstan
Lecturer Sep. 2020 – Jul. 2021
 - **Courses:** Algorithms and Data Structures, Advanced programming in C/C++, Web technologies.
 - Designed comprehensive course materials for 3 courses, including interactive lectures, assignments, and assessments
 - Coordinated and supervised student projects, guiding project development and ensuring successful completion.

- **Nazarbayev University AI Lab (#1 national institution in research)** Astana, Kazakhstan
Machine Learning Engineer *Jan. 2018 – Sep. 2019*
 - Engineered innovative AI solutions.
 - Enhanced machine learning models for face tracking and recognition on edge devices, achieving a 4% increase in predictive accuracy and a 15% reduction in inference time.
 - Designed generative deep learning models for face unmasking.
- **DAR** Astana, Kazakhstan
Backend Developer Intern *Jun.2020 – Aug. 2020*
 - Developed backend in Scala and Java

PUBLICATIONS

- **[ICIP2025]** R. Kairgeldin and M. A. Carreira-Perpiñán. Fast Image Vector Quantization Using Sparse Oblique Regression Trees. 2025 IEEE International Conference on Image Processing (ICIP), Anchorage, AK, USA, Sep. 14-17, 2025.
- **[NeSy2025]** R. Kairgeldin and M. A. Carreira-Perpiñán. Neurosymbolic models based on hybrids of convolutional neural networks and decision trees. 19th International Conference on Neurosymbolic Learning and Reasoning, Santa Cruz, CA, USA, Sep. 8-10, 2025.
- **[IAI Workshop at NeurIPS 2024]** R. Kairgeldin and M. A. Carreira-Perpiñán. Bivariate decision trees: Smaller, interpretable, more accurate. IAI Workshop at NeurIPS 2024, Vancouver, British Columbia, Canada, Dec. 15, 2024.
- **[UAI 2024]** R. Kairgeldin, M. Gabidolla, and M. A. Carreira-Perpiñán. Adaptive softmax trees for many-class classification. In Proc. of the 40th Conf. Uncertainty in Artificial Intelligence (UAI 2024), Barcelona, Spain, July 15–19 2024.
- **[KDD 2024]** R. Kairgeldin and M. A. Carreira-Perpiñán. Bivariate decision trees: Smaller, interpretable, more accurate. In Proc. of the 30th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (SIGKDD 2024), pages 1336–1347, Barcelona, Spain, Aug. 25–29 2024.
- **[BayLearn 2024]** R. Kairgeldin, M. Gabidolla, and M. A. Carreira-Perpiñán. Adaptive softmax trees for many-class classification. Extended abstract at Bay Area Machine Learning Symposium, 2024.
- **[BayLearn 2023]** R. Kairgeldin and M. A. Carreira-Perpiñán. Bivariate decision trees. Extended abstract at Bay Area Machine Learning Symposium, 2023.
- **[IEEE Transactions on Computational Social Systems]** M. Moradian, A. Dadlani, R. Kairgeldin and A. Khonsari, "Cost-Effective Activity Control of Asymptomatic Carriers in Layered Temporal Social Networks," in IEEE Transactions on Computational Social Systems, 2024

PROFESSIONAL ACTIVITIES

- **Reviewer:** The Conference on Uncertainty in Artificial Intelligence (UAI 2026).
- **Reviewer:** International Conference on Machine Learning (ICML 2025).
- **Reviewer:** Conference on Neural Information Processing Systems (NeurIPS), since 2024.
- **Reviewer:** Journal on Machine Learning Research Conference on Learning Representations (Journal of Machine Learning Research (JMLR)), since 2024.
- **Reviewer:** ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), since 2024.
- **Reviewer:** International Conference on Learning Representations (ICLR), since 2024.
- **Reviewer:** The international journal Advances in Data Analysis and Classification (ADAC), since 2022.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Matlab, Java, C#.
- **Frameworks:** pytorch, tensorflow, keras, scikit-learn, numpy, libsvm/liblinear, gurobi, scipy, etc.

LANGUAGES

English (fluent), Russian (native), Kazakh (fluent)